

day 06 | 250910 W

herein

Pascal's Triangle

First we will work on Pascal's triangle, since I realized that you don't know about nested loops yet. So we'll get that to work and talk about those.

A nested loop always has a structure like this:

```
for i in some_kind_of_range:
    for j in some_other_range_or_list:
        do_some_stuff_with_i_and_j
    do_more_stuff_with_i
```

What this does is loop through one list and at each step loop through another list. These can be related to each other as in our example in class, or can be totally separate.

Chapter 3: plotting!

We will also begin working on plotting things. First we will just cook up some "data" by which I mean we will plot a particular function, and then we will look at some options around how to make this look nice. There are MANY options on how to plot things and change different settings, but we will stick to the simplest ones. Here is an example of how to plot:

```
import matplotlib.pyplot as plt
import numpy as np

x = np.linspace(0,10)
y = x**2

fig0 = plt.figure()
ax0 = fig0.add_subplot()

ax0.plot(x, y, 'o', mfc='none')
```

This will plot $y = x^2$ from 0 to 10. Notice how I put in the option `'0'` and `mfc='none'` because that is the kind of plot that I like.

work on Pascal's Triangle

nested loops

In []:

```
In [1]: def factorial(x):
        f = 1
        if x<0:
            print('undefined')
        else:
            for i in range(1,x+1):
                f = f*i

            return(f)

        def binomial(n,k):
            if k>n:
                print('Learn more about Pascals Triangle')
            else:
                return(factorial(n)/factorial(k)/factorial(n-k))
```

In [12]: binomial(2,3)

Learn more about Pascals Triangle

```
In [15]: n = 1

        while n < 20:
            blist = []
            for k in range(n+1): # might change this to n+1
                b = binomial(n,k)
                blist.append(b)
            print(blist)

            n = n + 1
```

```

[1.0, 1.0]
[1.0, 2.0, 1.0]
[1.0, 3.0, 3.0, 1.0]
[1.0, 4.0, 6.0, 4.0, 1.0]
[1.0, 5.0, 10.0, 10.0, 5.0, 1.0]
[1.0, 6.0, 15.0, 20.0, 15.0, 6.0, 1.0]
[1.0, 7.0, 21.0, 35.0, 35.0, 21.0, 7.0, 1.0]
[1.0, 8.0, 28.0, 56.0, 70.0, 56.0, 28.0, 8.0, 1.0]
[1.0, 9.0, 36.0, 84.0, 126.0, 126.0, 84.0, 36.0, 9.0, 1.0]
[1.0, 10.0, 45.0, 120.0, 210.0, 252.0, 210.0, 120.0, 45.0, 10.0, 1.0]
[1.0, 11.0, 55.0, 165.0, 330.0, 462.0, 462.0, 330.0, 165.0, 55.0, 11.0, 1.0]
[1.0, 12.0, 66.0, 220.0, 495.0, 792.0, 924.0, 792.0, 495.0, 220.0, 66.0, 12.0, 1.0]
[1.0, 13.0, 78.0, 286.0, 715.0, 1287.0, 1716.0, 1716.0, 1287.0, 715.0, 286.0, 78.0, 13.0, 1.0]
[1.0, 14.0, 91.0, 364.0, 1001.0, 2002.0, 3003.0, 3432.0, 3003.0, 2002.0, 1001.0, 364.0, 91.0, 14.0, 1.0]
[1.0, 15.0, 105.0, 455.0, 1365.0, 3003.0, 5005.0, 6435.0, 6435.0, 5005.0, 3003.0, 1365.0, 455.0, 105.0, 15.0, 1.0]
[1.0, 16.0, 120.0, 560.0, 1820.0, 4368.0, 8008.0, 11440.0, 12870.0, 11440.0, 8008.0, 4368.0, 1820.0, 560.0, 120.0, 16.0, 1.0]
[1.0, 17.0, 136.0, 680.0, 2380.0, 6188.0, 12376.0, 19448.0, 24310.0, 24310.0, 19448.0, 12376.0, 6188.0, 2380.0, 680.0, 136.0, 17.0, 1.0]
[1.0, 18.0, 153.0, 816.0, 3060.0, 8568.0, 18564.0, 31824.0, 43758.0, 48620.0, 43758.0, 31824.0, 18564.0, 8568.0, 3060.0, 816.0, 153.0, 18.0, 1.0]
[1.0, 19.0, 171.0, 969.0, 3876.0, 11628.0, 27132.0, 50388.0, 75582.0, 92378.0, 92378.0, 75582.0, 50388.0, 27132.0, 11628.0, 3876.0, 969.0, 171.0, 19.0, 1.0]

```

plotting

```

In [16]: import numpy as np
import matplotlib.pyplot as plt

```

```

In [21]: x_data = np.linspace(0,10)
y_data = 5*x_data - 3

```

```

In [31]: fig0 = plt.figure()
ax0 = fig0.add_subplot()

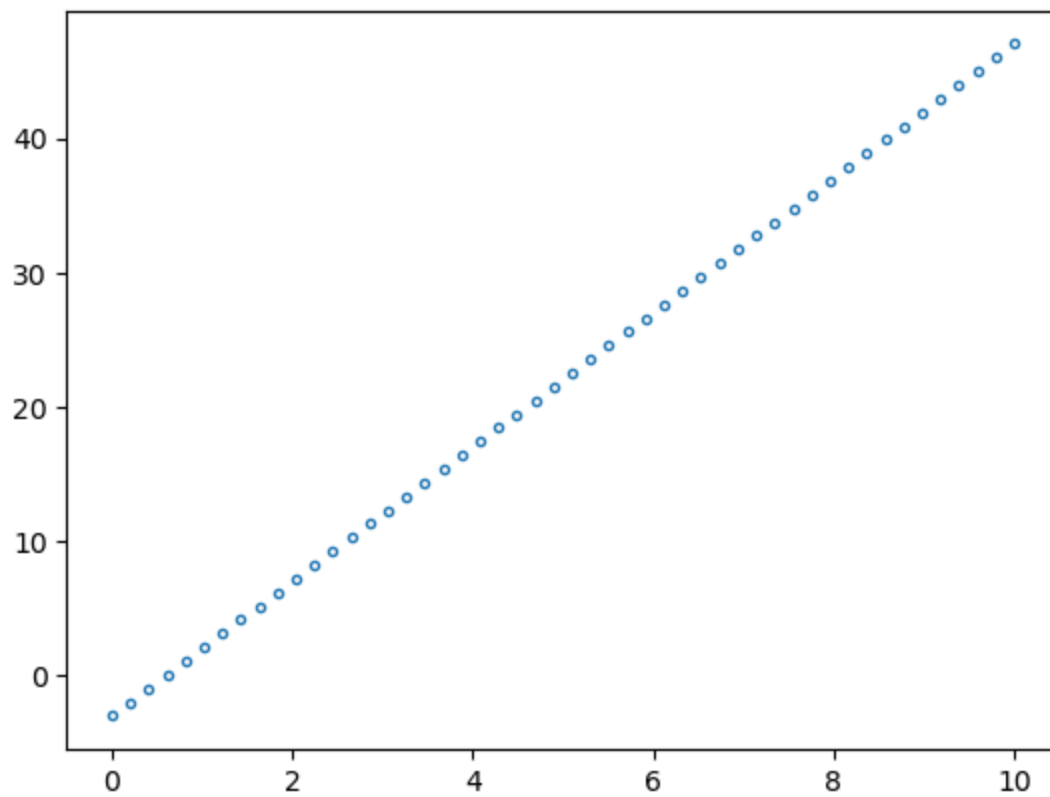
ax0.plot(x_data, y_data, '.', mfc="none")

```

```

Out[31]: [<matplotlib.lines.Line2D at 0x79099bb33110>]

```



In []: